

ShareLoop

Community Item Sharing Platform

Sprint 2 — Project Description

Version 2.0 | Sprint 2 | 2025...2026 Academic Year

1. Executive Summary

ShareLoop is a web-based community platform that enables local residents to share, exchange, and give away physical items with their neighbours. Rather than buying items they will only use once or twice, or discarding items that still have value, users can lend, swap, or give away possessions within a trusted local network.

The platform addresses a genuine social and economic problem: everyday items such as tools, baby clothes, textbooks, and sports equipment are purchased, used briefly, and left to gather dust — or thrown away. ShareLoop creates a circular economy at the neighbourhood level by connecting people who have items with people who need them, at no financial cost to either party.

ShareLoop is designed around three core exchange types: lending (temporary borrowing), swapping (permanent trade), and giving away (free gift). These three modes cover the vast majority of sharing needs within a community, and together they create a self-sustaining ecosystem of generosity and reciprocity.

Module Theme Alignment: ShareLoop directly embodies the module theme of Sharing, Exchange, and Building Community. Every feature — from the exchange types to the points system to the messaging layer — is designed to bring neighbours closer together and reduce unnecessary consumption.

2. Problem Statement

Modern urban and suburban life has created a paradox: people live in close proximity to hundreds of neighbours, yet they purchase items independently without knowing what already exists nearby. This leads to several compounding problems.

2.1 Financial Waste

Many items are bought for a single use — a power drill for one weekend project, a textbook for one semester, a dress for one event. The cost-of-living crisis has intensified this problem, particularly for students and young families who cannot easily absorb the cost of occasional-use items. Renting commercially is available for some items but is expensive, inconvenient, and rarely covers the everyday items most people actually need.

2.2 Resource Underutilisation

The same households that are buying new items already own items sitting idle in garages, attics, and storage units. Research into sharing economy behaviour consistently shows that items like power tools are used for an average of 12 to 15 minutes in their entire lifetime, yet occupy space and represent sunk costs for their owners. There is no friction-free, trusted way for these items to reach people who could benefit from them.

2.3 Environmental Impact

Every unnecessary purchase has an upstream environmental cost: raw materials extracted, energy used in manufacturing, packaging produced, and emissions generated by transportation. Items discarded to landfill before the end of their functional life compound this harm. A sharing platform directly reduces demand for new goods and extends the useful life of existing ones.

2.4 Social Isolation

Neighbours frequently do not know each other or what resources exist close by. The absence of a trusted, low-effort mechanism for local exchange means that even people who want to help their community lack an obvious way to do so. This contributes to isolation and reduces the social capital that makes communities resilient.

The Core Gap: There is no existing platform that combines local item sharing (lending, swapping, and giving), community trust-building, and ease of use in a single, free-to-access web application targeted at neighbourhood-level communities.

3. Our Solution: ShareLoop

ShareLoop solves these problems by providing a clean, intuitive web application where community members can list items they are willing to share and discover items available nearby. The platform is built on three principles: simplicity of use, community trust, and zero financial transactions.

3.1 The Three Exchange Types

ShareLoop is deliberately focused on physical item sharing only. All activity on the platform falls into one of three exchange types, each designed to serve a different community need.

Exchange Type	How It Works	Best For
Lending / Borrowing	Owner lists item as available to lend. Borrower requests it for an agreed period and returns it afterwards. Owner retains ownership throughout.	Tools, equipment, textbooks, camping gear, kitchen appliances, cameras
Swapping	Both parties list items and agree to a permanent trade. Each person walks away with the other's item. No money changes hands.	Baby clothes, children's toys, books, household goods, games

Giving Away	Owner lists item as a free gift with no expectation of return. Receiver simply collects it. A pure act of community generosity.	Outgrown clothes, old textbooks, unwanted gifts, surplus household items
-------------	---	--

3.2 Why These Three Types Cover the Full Spectrum

Lending handles temporary need without transferring ownership — ideal when the owner still wants the item back and the borrower only needs it briefly. Swapping handles the case where both parties have something the other wants, making it mutually beneficial and fair without any monetary valuation. Giving away handles decluttering and pure generosity, where the owner simply wants the item to go to someone who can use it. Together, these three types eliminate the need for any financial transaction on the platform.

4. Target Users and Personas

ShareLoop is designed for everyday community members, not tech-savvy early adopters. The platform must be accessible, trustworthy, and easy to use across a wide age range and varying levels of digital literacy.

4.1 Primary User Groups

User Group	Primary Needs & Motivations
Students	Need textbooks, calculators, sports equipment, and kitchen items on a limited budget. Benefit from borrowing expensive items for short periods. Example: “I need a graphing calculator for one exam — I’m not buying one for £100.”
Young Families	Children outgrow clothes, toys, and equipment rapidly. Swapping saves significant money and avoids waste. Example: “My toddler outgrew all his 12-month clothes — I need 18-month sizes and happy to pass these on.”
Local Residents	Need occasional-use tools and equipment (drills, ladders, garden tools) without buying them permanently. Benefit from meeting neighbours. Example: “I need a power drill for one weekend project — I’ll never use it again.”
Retirees	Have accumulated useful items they no longer need. Want to contribute to the community, stay connected, and declutter meaningfully. Example: “I have tools I no longer use and would love to lend them to younger neighbours.”

4.2 Secondary Users

Environmental advocates benefit from the platform's direct reduction of consumption and waste. Community organisations and mutual aid networks can use ShareLoop to coordinate resource sharing at scale. Universities may wish to promote ShareLoop to their student bodies as a financial support tool and sustainability initiative.

5. How ShareLoop Works

5.1 For Item Owners (Sharers)

1

1. List Your Item — Add a title, description, photos, condition rating, item category, and choose your exchange type (lend, swap, or giveaway). Set availability status.

2

2. Receive Requests — Get notified when community members request your item. View their profile, trust score, and any message they included.

3

3. Accept or Decline — Review the requester's ShareLoop profile including their tier, points, and ratings. Accept the right person for your item.

4

4. Coordinate via Messaging — Use in-app messaging to arrange pickup time, location, and any other details — no need to share personal contact information.

5

5. Complete the Exchange — Mark the item as collected or returned once the exchange is complete. Earn points and badges for your contribution.

6

6. Rate the Experience — Leave a star rating and optional review for the other person. Ratings build community trust over time.

5.2 For Item Seekers (Borrowers / Receivers)

1

1. Browse or Search — Search by keyword or browse by category and exchange type. Filter results to find exactly what you need.

2

2. View on Map — Switch to the map view to see what items are available near you, colour-coded by category.

3

3. Send a Request — Select the item you need, send a request with a brief message explaining your need and proposed dates (for lending).

4

4. Coordinate — Once the owner accepts, use in-app messaging to arrange pickup details safely within the platform.

5

5. Collect the Item — Meet the owner, collect the item, and complete the exchange. Return it by the agreed date if borrowing.

6

6. Rate Your Experience — Leave an honest star rating and review. Your rating also helps build your own trustworthiness on the platform.

6. Reward System: Points, Badges & Community Tiers

ShareLoop's reward system is central to the platform's long-term success. It is designed to encourage participation, reward generosity, build trust, and create a sense of community identity and progression. Unlike a purely financial incentive, the reward system is built on social recognition: users earn status and reputation, not money.

The system has three interconnected components: a Points System, a Badges System, and a Membership Tier System. These work together to give users short-term satisfaction (earning points and badges) and long-term motivation (progressing through membership tiers).

6.1 Points System

Points (called Loop Points) are the base currency of participation on ShareLoop. They are earned automatically when specific exchange milestones are completed. Points cannot be bought or transferred — they can only be earned through genuine community participation.

Action	Points	Notes
New account created	+50	Welcome bonus to encourage first activity
Complete your profile (photo, bio, location)	+20	One-time bonus
List your first item	+15	One-time bonus
Successfully lend an item (marked returned)	+10	Awarded to the lender
Successfully give away an item	+8	Awarded to the giver
Successfully complete a swap (both parties)	+20	Awarded to both parties
Receive a 5-star rating	+5	Per rating received
Leave a rating after an exchange	+3	Encourages feedback culture
7-day activity streak (login + browse)	+10	Awarded weekly for consistency
30-day streak	+25	Monthly consistency bonus
Refer a new user who joins	+15	Awarded when referral registers

Design Principle: Points are weighted to reward giving and lending more than receiving, and to reward consistent participation over time. This prevents a scenario where users only take from the community without contributing back.

6.2 Membership Tier System

Membership tiers give users a sense of progression and identity within the ShareLoop community. Each tier is unlocked by accumulating Loop Points. Higher tiers come with profile badges, increased visibility in listings, and early access to new platform features. Tiers reflect the depth of a user's contribution to the community.

Tier	Points Required	Perks & Recognition
Newcomer	0 – 99	Basic profile, can list and request items, access to all exchange types
Helper	100 – 299	Helper badge on profile, boosted visibility in search results, can leave extended reviews
Contributor	300 – 699	Contributor badge, priority listing placement, access to community statistics dashboard
Trusted Member	700 – 1499	Trusted Member badge (visually prominent), appear in “Trusted Near You” featured section, early access to beta features
Community Hero	1500+	Community Hero badge (gold), pinned listing priority, featured on homepage community spotlight, personal thank-you from ShareLoop team

6.3 Badges System

Badges are specific achievements earned for particular milestones and behaviours. Unlike tiers (which reflect total points accumulated), badges recognise specific types of contribution and can be displayed on a user's public profile. They add personality and specificity to a user's community identity.

Badge Name	How to Earn It	Category
First Loop	Complete your very first exchange (any type)	<i>Milestone</i>
Generous Soul	Give away 5 items for free	<i>Giving</i>
Tool Library	Lend 10 items successfully	<i>Lending</i>
Swap Master	Complete 5 successful swaps	<i>Swapping</i>
Five Stars	Receive ten 5-star ratings	<i>Reputation</i>
Perfect Streak	Maintain a 30-day activity streak	<i>Consistency</i>
Welcome Wagon	Refer 3 new users to ShareLoop	<i>Community Growth</i>
Neighbourhood Legend	Reach Community Hero tier	<i>Tier Achievement</i>
Quick Responder	Accept or decline all requests within 24 hours for a month	<i>Responsiveness</i>

Trusted Neighbour	Complete 20 exchanges with zero disputes	<i>Trust</i>
--------------------------	--	--------------

Why This System Works: Points + Tiers + Badges creates a three-layer motivation structure. Points give immediate feedback. Badges recognise specific behaviours and create collectible milestones. Tiers create long-term progression and community status. Together, they encourage generous behaviour without financial incentives.

6.4 Ratings and Trust Score

Separate from points and badges, every user has a Trust Score derived from peer ratings. After each exchange, both parties rate each other on a scale of 1 to 5 stars and can leave a written review. The Trust Score is the rolling average of all received ratings and is displayed prominently on every user's profile.

The Trust Score is designed to answer the key question any owner asks before lending a valuable item: can I trust this person? A high Trust Score — combined with a visible tier badge and exchange history — gives item owners the confidence to lend to people they have never met in person.

Trust Scores are visible on listing request notifications, search results, and user profiles. Owners are encouraged (but never forced) to favour higher-rated requesters. The last 10 written reviews are always publicly visible on a user's profile page.

7. Sprint 3 Core Features

Sprint 3 delivers the minimum viable product (MVP) of ShareLoop: a fully functional item sharing platform where users can register, list items, browse listings, send and manage requests, and rate each other. All data is stored in a MySQL database and all pages are rendered server-side using the PUG templating engine.

7.1 User Management

- User registration with email and password
- Email verification on account creation
- User login and session management
- User profile page displaying name, bio, profile photo, location, trust score, tier badge, and Loop Points
- Edit own profile (update bio, photo, and location)
- Community members list page (browse all registered users)
- View any user's public profile and their current listings

7.2 Item Listings

- Create a new listing with title, description, up to 5 photos, category, condition, and exchange type
- Categories: Tools, Books, Baby & Kids, Sports, Household, Electronics, Clothing, Other
- Conditions: Like New, Good, Fair, Well-Used
- Exchange types: Lend, Swap, Give Away
- Set item availability status (Available / Unavailable)
- Edit and delete own listings
- Browse all available listings (paginated)
- Search listings by keyword
- Filter listings by category and exchange type
- Detailed listing page showing full description, photos, owner info, and request button

7.3 Request System

- Send a request to borrow, swap, or receive an item
- Include a message to the owner explaining your need
- View all requests you have sent (with current status: pending, accepted, declined, completed)
- View all requests received on your own listings
- Accept or decline incoming requests
- Mark an item as returned or exchange as completed
- Automatic Loop Points awarded when an exchange is marked complete

7.4 Required Pages

Page	Purpose
Homepage	Welcome page explaining ShareLoop, how it works, and call to action to register or browse
Users List Page	Show all community members with tier badge, trust score, and link to profile (required by brief)
User Profile Page	Display user details, tier, points, badges, trust score, and their active listings (required by brief)
Listings Page	Browse all available items with search and filter controls (required by brief)
Listing Detail Page	Full information about a specific item including photos, description, owner info, and request button (required by brief)
Categories Page	Browse items organised by category with item counts (required by brief)
Create / Edit Listing	Form to add or edit a listing with photo upload
Dashboard	Personal overview: my listings, my requests sent, requests received on my items
Request Management	Accept, decline, and complete requests with messaging thread
Login / Register	Authentication pages with email verification flow

8. Sprint 4 Advanced Features

Sprint 4 builds on the MVP by adding four significant features that substantially improve the platform's usefulness and community-building capability. These features move ShareLoop from a functional MVP to a compelling, market-ready application.

8.1 Google Maps Integration

A map view is added to the listings page showing all available items as colour-coded pins. Users can visually see what is available near them without having to browse individual listings. Clicking a pin shows a preview card for the item with a link to the full listing.

- Map view with colour-coded pins by category
- Filter map by distance (1km, 5km, 10km from user location)
- Near Me functionality using browser geolocation
- Distance displayed on all listings (e.g. "0.8km away")
- Geocoding of user location (address to latitude/longitude)
- Haversine formula for accurate distance calculation

8.2 In-App Messaging

Users can send and receive direct messages within ShareLoop, linked to specific items and requests. This removes the need to share personal contact details and keeps all exchange coordination within the platform, creating a safer, auditable communication record.

- Direct messaging between any two users
- Conversations linked to specific item requests
- Full message history preserved
- Unread message indicator on the navigation bar
- Clean, minimal chat interface

8.3 Enhanced Reward System (Sprint 4 Expansion)

The points, badges, and tier system described in Section 6 is fully implemented in Sprint 4. Sprint 3 handles automatic point awarding for completed exchanges; Sprint 4 adds the full badge detection logic, tier upgrade notifications, and the community leaderboard.

- Full badge detection and award logic running server-side
- Tier upgrade notification (in-app toast message when user advances)
- Community leaderboard showing top Loop Points earners
- Badge showcase on public user profiles
- Points history log available in user dashboard

8.4 Item Recommendation Engine

A lightweight recommendation algorithm surfaces relevant items to users based on their behaviour on the platform. Recommendations appear on the user's dashboard and on the listings page as a "You might also like" section.

Signal	Weight	Description
User History Match	40%	Categories the user has previously requested or browsed
Distance Score	30%	Items within 5km are prioritised heavily
Item Popularity	20%	Items that have received the most requests recently
Recency	10%	Newer listings ranked above older ones at the same score

9. Technical Architecture

ShareLoop is a server-rendered web application built on Node.js with Express. The PUG templating engine handles all HTML rendering. MySQL 8.0 is the primary data store. The entire application is containerised with Docker for consistent development and deployment environments.

9.1 Technology Stack

Layer	Technology	Purpose
Frontend	HTML, CSS, JavaScript	UI, responsive design, map rendering
Templating	PUG (Jade)	Server-side HTML rendering
Backend	Node.js + Express.js	RESTful routing, business logic, session management
Database	MySQL 8.0	Relational data storage with geospatial support
Auth	bcrypt + express-session	Password hashing and session-based authentication
File Uploads	Multer	Handling photo uploads for listings and profiles
Maps	Google Maps JavaScript API	Geolocation, geocoding, and map rendering (Sprint 4)
DevOps	Docker + docker-compose	Containerised app and database for consistent environments
CI/CD	GitHub Actions	Automated testing, linting, or build verification on push

9.2 Project Scope Boundaries

What ShareLoop IS Building	What ShareLoop is NOT Building
✓ Item sharing platform (lending, swapping, giving)	✗ Skills or services sharing (removed per feedback)
✓ Three clear exchange types with distinct flows	✗ Payment processing (money-free platform)
✓ Location-based discovery with Google Maps	✗ Delivery or shipping (local pickup only)
✓ In-app messaging for exchange coordination	✗ Insurance or damage liability system

✓ Trust system via ratings and reviews	✗ Advanced calendar or booking system
✓ Points, badges, and tiered membership rewards	✗ Mobile app (responsive web only)
✓ Smart item recommendations	✗ Commercial advertising or paid promotion

10. Example User Journeys

The following journeys illustrate how real users would interact with ShareLoop in practice, demonstrating the value of each exchange type and the rewards system.

Journey 1: Sarah Borrows a Textbook

User Profile: Sarah, 19, first-year university student. Needs a statistics textbook for exam prep but cannot justify buying one for £45.

- Day 1 — Discovery: Sarah searches “statistics textbook”. Finds Marcus, 0.8km away, lending his. His profile shows Contributor tier, 4.7 stars, 12 successful exchanges.
- Day 1 — Request: Sarah sends a request with a message: “Hi! I need this for exam prep next month — I can collect any time this week and will take great care of it.”
- Day 2 — Accepted: Marcus accepts. They use in-app messaging to arrange a library pickup for Tuesday at 3pm.
- Week 5 — Return: Sarah returns the book after her exam. Both rate 5 stars and leave reviews.
- Outcome: Sarah earns +3 points (for rating). Marcus earns +10 points (lending) and +5 points (5-star rating). Sarah saved £45. Marcus has now completed 13 exchanges and is approaching Trusted Member tier.

Journey 2: Maya Swaps Baby Clothes

User Profile: Maya, 31, mum of a toddler. Has a bundle of 6-12 month baby clothes her son has outgrown. Needs 18-24 month clothes.

- Week 1 — Listing: Maya lists a bundle of 20 baby clothes (Like New condition, swap exchange type). She notes she is looking for 18–24 month clothes in return.
- Week 1 — Requests: She receives 3 requests. She chooses Emma, who lives 1.2km away, has exactly the clothes Maya needs, and has a Helper tier badge with 4.8 stars.
- Week 2 — Coordinate: They message to agree a swap at the local park. Both bring their kids.
- Week 2 — Exchange: Both are happy. Both mark the exchange as complete.
- Outcome: Both earn +20 Loop Points for the swap. Both leave 5-star ratings. Both earn +3 more points for rating. Maya unlocks the “Swap Master” badge progress. Both families saved over £50 each.

Journey 3: David Gives Away Tools

User Profile: David, 67, retired. Has accumulated tools he no longer uses. Wants to declutter and give back to his community.

- Month 1 — Listing: David lists 5 tools as giveaways: a hand drill, a level, tape measures, a toolbox, and a set of chisels.
- Month 1 — Requests: Multiple students request items. David reads their messages and chooses those who explain a genuine need (new accommodation, first DIY project).
- Month 2 — Giveaways: David arranges pickups for each item. He earns +8 Loop Points per giveaway (40 total) plus +5 for each 5-star rating received.
- Month 3 — Recognition: David now has 290 points (approaching Contributor tier), a 5.0 Trust Score, 8 written reviews, and has unlocked the “Generous Soul” badge. He is featured in the “Trusted Near You” section.
- Outcome: David feels valued and purposeful. Students saved money on essential items. Community is stronger. David is now ShareLoop’s most active member in his postcode area.

11. Success Criteria

Sprint 3 Success

- Users can register, verify email, and create a profile
- Items can be listed with photos, descriptions, and all required metadata
- Users can browse, search, and filter items across all categories
- Request system is fully functional (send, accept, decline, complete)
- All required pages specified in the brief are functional and styled
- All data is persisted in and retrieved from MySQL
- Loop Points are automatically awarded when exchanges are marked complete

Sprint 4 Success

- Map view displays all available items with accurate pins within 10km of the user
- Users can message each other within the platform and receive unread indicators
- Full badge and tier logic runs server-side and updates profiles in real-time
- Community leaderboard shows top contributors by Loop Points
- Recommendation engine surfaces relevant items on the dashboard
- At least one GitHub Action is running successfully on every push

Overall Platform Success

- Platform is usable and intuitive for non-technical users across all age groups
- Exchanges complete successfully from request to rating
- Trust is demonstrably built through the ratings and tier system
- The three exchange types are clearly differentiated and well-understood by users

- The community grows organically through referrals and word of mouth

12. Alignment with Module Theme

ShareLoop was designed from the ground up to embody the three pillars of the module theme: Sharing, Exchange, and Building Community.

Pillar	How ShareLoop Delivers It
Sharing	Physical items are shared within the community rather than hoarded or discarded. The platform makes sharing the path of least resistance — easier than selling, simpler than throwing away, and more rewarding than leaving items idle.
Exchange	Three distinct exchange types serve different needs: lending creates reciprocal obligation, swapping creates mutual benefit, giving away creates one-way generosity. No money changes hands; the currency is trust and community reputation.
Building Community	Every exchange is a connection between neighbours. Ratings and reviews create a web of trust. Tiers and badges create identity and pride. The map makes the community tangible and visible. ShareLoop turns strangers into neighbours.

ShareLoop — Share More. Waste Less. Connect Better.

Sprint 2 Project Description | 2025–2026